

Emergency Department + EMS + Inpatient Simulation — Functional Specification

1) Purpose & Scope

A discrete-event simulation (SimPy) that represents a hospital's patient flow end-to-end:

- **ED Acute Areas** (e.g., A/B sides) where patients stay in-bed until disposition.
 - **Arrivals** from **Walk-ins** and **EMS** as separate stochastic streams.
 - **Acuity-based** ED bed allocation (not FIFO).
 - **Doctors** assigned to areas with **front-loaded signup quotas** and **panel limits**.
 - **Nurses** execute **procedures, labs, diagnostics (DI)** using **ratio** or **team** workload models.
 - **Site capability toggles** (e.g., CT present/absent, transfer-only admissions).
 - **External DI** and **Admit Transfers** if services or admitting aren't available.
 - **Inpatient Units** (Medicine, Surgery, ICU, etc.) with capacity and LOS; ED boarding when units are full.
 - **Rich event logging** and **patient summaries** for KPIs and calibration.
 - Built as a **single-site** engine with an optional hook to **receive EMS calls from a zone dispatcher**.
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2) Time & Units

- **Clock:** minutes from simulation start.
 - **Horizon:** configurable hours for walk-ins, EMS, and direct inpatient admits.
 - **Durations:** all waits, services, buffers, offload, boarding, and LOS in minutes.
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3) Entities & Roles

Patients

- **Modes:** Walk-in, EMS (critical/non-critical), or Direct Inpatient Admit.
- **Triage features:** age, temperature, O₂ saturation, blood pressure, GCS, complaint code (1–168), trauma flag, mental-health flag.
- **CTAS** (1–5) for EMS; drawn from a configurable mix.
- **Acuity score:** weighted function of vitals/flags/complaint/age + CTAS bonus; **higher** = **higher priority** for ED beds.

- **LWBS threshold:** maximum personal wait-to-bed for walk-ins/EMS not yet bedded.

Lifecycle (ED branch): Waiting → **Bed** → Assessment(s) → Orders (round 1, optional round 2) → Stabilization → Consult → **Disposition** (Discharge or Admit/Transfer).

Lifecycle (Inpatient): Waiting → **Unit Bed** → **LOS** → Discharge.

ED Areas & Beds

- Each area (e.g., **A**, **B**) has a fixed number of beds.
- A **priority waiting list** per area is always **sorted by acuity** (tie → earlier arrival).
- Patients **remain in their bed** until disposition (no bouncing back to waiting).

Doctors (per Area)

- **Shift window:** start time + length (midnight wrap supported).
- **Hourly signup quotas:** max **new** patients the doctor can take each hour (front-loaded patterns allowed).
- **Max active panel:** cap on concurrent under-care patients.
- **Times:** separate draws for **assessment** and **reassessment**.
- **Behavior:** takes **new bedded** patients first (if eligible), otherwise **reassesses** patients with completed orders. Serial work—one patient at a time.

Nurses (Orders Execution)

Two models per area:

- **Ratio model:** e.g., 1 nurse per 2 beds; **each bed has its own FIFO** of orders.
- **Team model:** **shared FIFO** across a cluster of beds; multiple nurses pull next order.
- **Lab support toggle:** if on, nurse time for labs is minimal (collection only) and lab **turnaround** can be modeled as a separate delay.

Offload Nurses (EMS)

- **Per-hour schedule** defines how many offloads can run in parallel (token pool).
- Each offload consumes one token for the **offload service time**.
- **Crew hospital time:** fixed time crews remain tied up, **independent of offload duration** (paperwork/equipment reset).

Inpatient Units

- Each unit (e.g., Medicine, Surgery, ICU) has **capacity** and a **LOS distribution**.
 - **Admission queues** are FIFO.
 - **ED admits** board in ED until a unit bed is available; **direct admits** join unit queues directly.
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4) Arrival Streams & Triage

Walk-ins

- **Poisson** arrivals per hour (configurable by hour).
- Routed to an ED area (random or site rule).
- Immediate triage → added to that area's **acuity** waiting list.

EMS (Single-Site Mode)

Two sources:

1. **Internal generation:** EMS arrivals are **Poisson** by hour.
2. **External hook:** `receive_ems_call(area_hint, ctas, critical)` allows a **zone dispatcher** to inject calls.

On EMS arrival:

- Assign **CTAS** from configured mix.
- **Critical flag** with probability $p_{critical}$ (e.g., ~1%).
- Join **EMS FIFO** (preserves ambulance order) before offload.

Direct Inpatient Admits (Optional)

- **Poisson** by unit; join **unit** admission queue (do not go through ED).
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5) EMS Pipeline (Detailed)

Goal: realistically constrain the moment of hospital takeover from ambulance to ED.

1. EMS FIFO (Bay Queue).

Ambulances line up by arrival time.

2. Offload Nurse Gate (Token Pool).

When a token is available (per-hour schedule), offload starts. Offload **service time** is drawn and represents transfer of care + immediate paperwork.

3. Path Selection (Logged).

- **Critical:** Must wait until **ED bed is free and a doctor can accept** (on shift, under hourly quota, and under panel cap) → **direct-to-bed**.
- **Non-critical:**
 - With probability **p_direct_to_bed** and if bed+doctor available → **direct-to-bed**.
 - Else if **download/staging** has capacity → park there; later pulled to ED bed by acuity.

- Else → **ED waiting room** (acuity queue like walk-ins).

4. Crew Hospital Time (Independent Lockout).

Regardless of offload duration, crews remain unavailable for a **fixed hospital time**; this reflects real minimums for crew reset. We log **ems_crew_clear** separately.

Logged events: **ems_arrival**, **ems_offload_start**, **ems_offload_end** (path), **ems_crew_clear**.

6) ED Bed Allocation (Priority Logic)

Continuous loop per area:

- **LWBS pass:** any waiting patient whose (now – arrival) > personal LWBS threshold is marked **LWBS** and removed.
 - If a bed is free and the waiting list is non-empty:
Select highest acuity (tie → earlier arrival) → **assign bed** → start in-bed flow.
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7) Doctor–Patient Interaction

Eligibility to take a new patient:

Doctor must be **on shift**, **below hourly signup quota**, and **below panel cap**.

Selection priority:

1. **New bedded** patients not yet assessed (sorted by **higher acuity**, then **earlier arrival**).
2. **Reassessment** candidates (orders completed; same sort).

Work units:

- **Assessment:** duration draw; may place **Orders Round 1** (procedures, labs, and possibly DI).
 - **Reassessment(s):** duration draw; can trigger **Orders Round 2** (no DI by design).
 - Doctors work **serially** (one at a time). First contact with a new patient logs a **signup** and counts against **hourly quota** and **panel**.
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8) Orders Lifecycle (Procedures, Labs, DI)

Generation:

After assessment/reassessment, **Bernoulli draws** decide which orders are placed:

- **Procedures (proc):** probability + nurse service time draw + **work score** (1–10) for analytics.

- **Labs:** probability + nurse time (minimal if **lab support** is on) + optional **lab turnaround delay**.
- **Diagnostics (DI):** probability (round 1 only) + modality choice (**X-ray / CT / US**) + nurse handling time.

Site capability check:

If a modality is **unavailable** (e.g., no CT), we log **transfer_needed** and simulate **external DI** as a **single round-trip time block** (travel + study + return). After it completes, DI is considered done.

Queuing & Execution (Nursing):

- **Ratio model:** order goes to the **bed's FIFO**; assigned nurse cycles through their bed group.
- **Team model:** order goes to **shared FIFO**; any team nurse can take next order.
- We log `order_enqueued`, `order_start`, `order_done` (with modality where applicable).

Completion:

- **Round 1** completes when all its orders (including any external DI block) finish.
- **Round 2** completes when its orders finish (no DI in round 2).

9) Stabilization, Consults, and Disposition

When all required orders are complete:

1. **Stabilization buffer** (short, keeps ED bed).
2. **Consult (probabilistic):**
 - Pick a service (1..N), attend after a short delay.
 - **Admit probability by service** decides path:
 - **Admit:**
 - **Full-service site:** start **boarding** in ED until an inpatient bed opens; then transfer to unit (ED bed frees).
 - **Transfer-only site:** simulate **admit transfer out**; ED bed held until transfer completes.
 - **Discharge:** apply a **post-discharge buffer** (paperwork/observation); then ED bed frees.

Logged: `stabilization_start`, `consult_requested`, `consult_attended`, `boarding_start`, `inpatient_transfer` or `admit_transfer_*`, `discharge_buffer_start`, `discharged`.

10) Inpatient Units (Detailed)

Structure: each unit has **capacity**, **LOS distribution**, and a **FIFO admission queue**.

Admission sources:

- **ED admits:** if no bed, the patient **boards in ED**; admitted to unit when capacity frees.
- **Direct admits:** arrive by a Poisson process to the **unit queue** (bypass ED).

Bed turnover:

- On **unit admission**, LOS is drawn; after LOS completes, the **unit bed frees**, and the next queued patient is admitted immediately.

Logged: inpatient_admit_request, inpatient_admitted, inpatient_discharged.

11) Queue Disciplines (At a Glance)

- **EMS offload:** **FIFO** by ambulance arrival (offload nurse gate).
 - **ED waiting list:** **Acuity priority**, then arrival time.
 - **Nurse orders:** **FIFO** (per bed in ratio model, shared **FIFO** in team model).
 - **Inpatient admissions:** **FIFO** per unit.
 - **Doctor selection:** **New first** (if eligible), else reassess; both **higher acuity** then **earlier arrival**.
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12) Randomness & Distributions

- **Arrivals:** Poisson (walk-in hourly, EMS hourly, direct admits by unit/hour).
- **CTAS mix:** categorical probabilities.
- **Doctor times:** assessment & reassessment draws.
- **Orders:** Bernoulli flags per type; DI modality mix; nurse service time draws; work score (1–10).
- **Lab:** if lab support on, nurse time minimal; lab TAT can be modeled separately.
- **External DI:** single round-trip time block (travel + study + return).
- **Consults:** request probability; **service-specific** admit probability.
- **Buffers:** stabilization and post-discharge.
- **Admit transfers:** time block (transfer-only sites).
- **LOS:** per inpatient unit (Medicine, Surgery, ICU, etc.).
- **LWBS:** per-patient thresholds.

- **Offload:** offload nurse capacity per hour + offload service time + **crew hospital time**.
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13) Outputs

Patient Table (one row per patient)

- **Descriptors:** pid, mode, area/unit, CTAS, acuity, complaint.
- **Access/flow times:** arrival, bed-in, first doctor, order round completion(s), disposition.
- **Outcomes:** LWBS, admitted flag, unit (if admitted), ED LOS, unit LOS.
- **Path flags:** external DI, transfer out, boarded in ED, etc.

Event Table (one row per event)

- Timestamps + context: arrivals, queued_for_bed, LWBS, bed_in/out, doctor signup/assess start/done, order enqueue/start/done, external DI start/done, consults, stabilization, boarding, unit admit/discharge, **EMS** offload start/end + **crew clear**.

KPIs you can compute:

- **Access:** Arrival → Bed, EMS offload delay, EMS crew lockout.
 - **Flow:** Bed → First Doctor, Orders (round 1 & 2) durations, time to reassessment, disposition time.
 - **Outcome:** ED LOS by mode/CTAS/area; LWBS rate/drivers.
 - **Capacity:** ED bed occupancy, boarding time, unit occupancy, staging occupancy, nurse queue depth.
 - **Productivity:** orders per nurse/hour, doctor panel sizes & signups vs quotas.
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14) Assumptions (Explicit)

1. **Poisson arrivals** capture peaks/troughs via hourly rates.
2. **Acuity score** governs ED bed allocation; **EMS FIFO** governs offload order only.
3. **Doctors** are serial, obey **shift**, **hourly signup quotas**, and **panel caps**.
4. **Orders round 1** may include DI; **round 2 excludes DI** (by design).
5. Missing modalities trigger **external DI** as a **single time block** (round-trip approximation).
6. **Nursing** executes orders in FIFO at the chosen structure; **work score** is analytic only (not a capacity constraint).
7. **Offload nurses** and **crew hospital time** are the only pre-ED bottlenecks for EMS handover.
8. **Transfer-only admissions** keep ED bed until transfer completes.
9. **Boarding** is explicit when inpatient capacity is full.

10. **No diversion** yet (sites do not refuse EMS); easy to add as a rule.

15) Configuration Knobs (Most Used)

- **Arrivals:** walk-in hourly lambdas; EMS hourly lambdas & CTAS mix; *p_critical*; *p_direct_to_bed*; direct admit lambdas per unit.
 - **ED Areas:** bed counts; nurse model (ratio/team); ratio size or team nurse count; **lab_support**.
 - **Doctors:** shift windows; per-hour signup quotas; max panel; assess/reassess time draws.
 - **Orders:** proc/lab/DI probabilities; DI modality mix; service time draws; work score draws.
 - **Capabilities:** has_CT / has_Xray / has_US; transfer_only_admit; external DI/admit transfer time draws.
 - **EMS:** offload nurses per hour; staging capacity; offload service time; **crew hospital time**.
 - **Inpatient Units:** capacities; LOS distributions; mapping from ED consult services to units; direct admit rates.
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16) Validation & Scenario Design

- **Face validity:** Arrival → Bed (by CTAS), Offload delay, Boarding time, ED LOS, LWBS vs site baselines.
 - **Stress tests:** surge arrivals; fewer offload nurses; no CT; transfer-only admits.
 - **Sensitivity:** doctor quotas/panel caps; nurse ratio vs team; crew hospital time; unit capacities/LOS.
 - **Policy trials:** prioritize certain CTAS for direct-to-bed; adjust consult/admit probabilities by service.
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17) Known Limitations / Roadmap

- **External DI** and **admit transfers** are single time blocks (no explicit receiving-site queues yet).
 - **Lab TAT** simplified when lab support is on; can add per-test TATs.
 - **Reassessment targeting** for doctors is heuristic (ready = no pending orders).
 - **Diversion & zone routing** not in single-site (hooks exist for future multi-site).
 - **Utilization** is inferred from events; explicit state-based utilization tracking could be added.
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18) Practical Tuning Checklist

- **Acuity fairness:** If CTAS 1–2 aren't consistently at the top, increase CTAS bonus and/or vitals weighting.
- **LWBS too high:** Raise bed counts or lower LWBS threshold; improve acuity prioritization.
- **First-doc too slow:** Lower hourly signups or panel caps (or add a doctor); reduce assess times.
- **Orders backlog:** Increase nurse capacity (ratio/team), shorten order times, or turn on **lab_support** + model TAT.
- **Offload jammed:** Add offload nurses, shorten offload service time, or lower **p_direct_to_bed**; if crews clear too fast/slow, adjust **crew hospital time** only.
- **Boarding dominates:** Increase unit capacity, reduce unit LOS, or reduce admit probability by service.

Performance Indicators Glossary

How to read these

- **Event fields** refer to the event log (e.g., `arrival`, `bed_in`, `doc_assess_start`, `ems_offload_start`).
 - **Per-patient times** are typically computed as the **first** timestamp of the event type for that patient.
 - Always report **median (p50)** and **tail (p90/p95)**, and segment by **mode (Walk-in/EMS)**, **CTAS**, **area**, and **hour of day**.
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Access & Intake

Arrival → Bed Time

Time from patient arrival to occupying an ED bed.

Formula: `bed_in` - `arrival`.

Why it matters: Front-door access; strong driver of LWBS.

Arrival → First Doctor Contact

Time from arrival to the first assessment start.

Formula: `doc_assess_start(first)` - `arrival`.

Why: Clinical access and risk exposure.

Arrival → Triage Complete *(if you log it)*

Time to triage completion.

Formula: `triage_done` - `arrival`.

Why: Intake responsiveness; useful when triage is a visible bottleneck.

LWBS Rate

% of arrivals who leave without being seen (no `bed_in` before LWBS).

Formula: `count(lwbs)` / `count(arrivals)`.

EMS-Specific

EMS Offload Delay

Ambulance arrival to offload start.

Formula: `ems_offload_start` - `ems_arrival`.

Purpose: Nurse gate bottleneck.

Offload Service Duration

Offload start to offload end.

Formula: `ems_offload_end` - `ems_offload_start`.

Purpose: Bedside handover & paperwork efficiency.

Crew Hospital Lockout

Independent minimum crew time at hospital.

Formula: `ems_crew_clear - ems_offload_start`.

Note: Not the same as offload duration.

Direct-to-Bed Rate (EMS)

% EMS patients routed directly to a bed.

Formula: `count(ems_offload_end.path ∈ {direct_bed, direct_bed_critical}) / count(ems_offload_end)`.

Staging Utilization (Download Area)

Occupancy of staging over time; or % EMS routed to staging.

Formula (rate): `count(ems_offload_end.path == download) / count(ems_offload_end)`.

ED Flow

Bed → First Doctor Contact

Time from bed assignment to first assessment.

Formula: `doc_assess_start(first) - bed_in`.

Why: Physician latency after bedded.

Assessment Duration (Initial / Reassess)

Doctor time per session.

Formulas:

- Initial: `doc_assess_done(first) - doc_assess_start(first)`
- Reassess: for each reassess pair, `done - start`.

Orders Round 1 Duration

From first round placement to completion of all R1 orders (incl. external DI time block).

Formula: `orders_first_round_done - doc_assess_done(first)`.

Orders Round 2 Duration (*if placed*)

Formula: `orders_second_round_done - doc_assess_done(reassess_that_placed_R2)`.

Stabilization Time

Post-orders buffer prior to disposition.

Formula: `discharge_buffer_start` or `boarding_start - stabilization_start`.

Disposition & Outcomes

ED Length of Stay (ED LOS)

Arrival to final ED disposition (discharge OR inpatient transfer start/complete).

Formula:

- Discharge path: `discharged - arrival`
- Admit/transfer path: `inpatient_transfer` or `admit_transfer_complete - arrival`.

Boarding Time (ED)

Time holding an ED bed after decision to admit until inpatient unit accepts.

Typical Formula: `inpatient_transfer - boarding_start`.

Why: Captures inpatient constraint impact on ED.

Admit Rate (by Service/CTAS/Mode)

% of ED encounters admitted or transferred for inpatient care.

Formula: `count(admitted) / count(ED encounters)`.

External DI Utilization

% of encounters that required external diagnostics due to local capability gaps.

Formula: `count(external_di_start unique pids) / count(ED encounters)`.

Capacity & Queues

ED Bed Occupancy

Average % of ED beds occupied (by area).

Formula: time-weighted occupancy from `bed_in/bed_out` events / total beds.

Inpatient Unit Occupancy (*if unit model enabled*)

Average % of unit beds occupied.

Formula: similar time-weighted approach using unit admit/discharge events.

Waiting List Depth (ED)

Average and max number of waiting patients by area.

Source: sampled queue length over time or reconstructed from `queued_for_bed` and `bed_in`.

Nurse Queue Depth

Average and max orders waiting (bed FIFO or shared FIFO).

Source: order enqueue/start/done events or internal queue probes.

Productivity & Workload

Doctor Signups per Hour

New patients assigned per doctor per hour.

Formula: count of `doc_signup` events, binned hourly.

Active Panel Size (Doctor Load)

Concurrent patients under a doctor's care over time.

Source: track `doc_signup` (+1) and final disposition/release (−1).

Orders per Nurse per Hour

Completed orders binned by nurse/hour.

Formula: count of `order_done` grouped by nurse and hour.

Order Mix & Effort

Distribution of procedure/lab/DI counts and **sum of work scores** per encounter or per hour.

Quality & Safety Proxies

High-Acuity Waits

Arrival → Bed for CTAS 1–2 (median and p90).

Purpose: Safety of the sickest.

Return Within X Hours *(if modeled later)*

% discharged patients who re-present within X hours.

(Placeholder for future data integration.)

Throughput / Service Totals

Total Arrivals (by Mode)

Walk-in vs EMS counts over horizon.

Total Dispositions

Discharged, Admitted (local), Transfer Out (admit transfer).

External DI Count

Number of external DI episodes (by modality).

Calibration Aids

Percentiles

Always report **p50, p75, p90, p95** for time-based KPIs.

Segmentation

Slice by **CTAS**, **mode (Walk-in/EMS)**, **area (A/B)**, **hour of day**, and **day part (e.g., 4-hour bins)**.

Targets & Alerts

- Arrival → Bed p90 (CTAS 1–2)
- Offload delay p90
- Boarding time median

- LWBS rate
Use these as operational “red lines” for scenarios.
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Common Pitfalls & Notes

- **Arrival timestamps:** Use the **first** arrival event per patient; EMS can have both `arrival` and `ems_arrival`—treat `ems_arrival` as the intake time for EMS mode.
- **Multiple assessments:** Use the **first** `doc_assess_start` as “first doctor.”
- **Orders completion:** Round 1 is not done until **all** its orders (and any external DI block) complete.
- **Disposition choice:** Prefer `discharged`, `inpatient_transfer`, or `admit_transfer_complete` as the terminal event (not `bed_out` alone).
- **Crew clear vs offload:** Don’t confuse **crew lockout** with **offload duration**—they’re logged separately.
- **Boarding:** Start with `boarding_start`, end with `inpatient_transfer`; if transfer-only, use `admit_transfer_complete`.